From Orthogonal to Multiple Orthogonal Polynomials

(and back)



Maksim Kosmakov

University of Toledo

Joint work with Roozbeh Gharakhloo and Kenta Miyahara

October 30, 2026

Orthogonality with respect to one weight

Definition (Orthogonal polynomials on the real line)

Let μ be a positive measure on $\mathbb{R}$ with finite moments $m_k = \int_{\mathbb{R}} x^k \, \mathrm{d}\mu(x)$ for $k \geq 0$. The nth *orthogonal polynomial* is the unique monic polynomial $p_n(x) = x^n + \cdots$ satisfying

$$\int_{\mathbb{R}} p_n(x) x^k \, \mathrm{d}\mu(x) = \delta_{nk} h_n^2, \qquad k = 0, \ldots, n,$$

where $h_n^2 = \int_{\mathbb{R}} p_n(x)^2 \, \mathrm{d}\mu(x) > 0$.

Family	Measure $\mathrm{d}\mu(x)$	Support
Hermite	$e^{-x^2/2} \, \mathrm{d}x$	$\mathbb{R}$
Laguerre	$x^\alpha e^{-x} \, \mathrm{d}x$, $\alpha > -1$	$(0, \infty)$
Jacobi	$(1-x)^\alpha (1+x)^\beta \, \mathrm{d}x$, $\alpha, \beta > -1$	$(-1, 1)$
$\vdots$	$\vdots$	$\vdots$



Orthogonal polynomials from Hankel determinants

The *Hankel determinants* of the moments of μ are

$$D_n = \det \begin{pmatrix} m_0 & m_1 & \cdots & m_{n-1} \\ m_1 & m_2 & \cdots & m_n \\ \vdots & \vdots & & \vdots \\ m_{n-1} & m_n & \cdots & m_{2n-2} \end{pmatrix}, \qquad m_k = \int_{\mathbb{R}} x^k \, \mathrm{d}\mu(x).$$

The monic OP can then be expressed in terms of these moments:

$$p_n(x) = \frac{1}{D_n} \det \begin{pmatrix} m_0 & m_1 & \cdots & m_{n-1} & 1 \\ m_1 & m_2 & \cdots & m_n & x \\ \vdots & \vdots & & \vdots & \vdots \\ m_n & m_{n+1} & \cdots & m_{2n-1} & x^n \end{pmatrix}, \qquad h_n^2 = \frac{D_{n+1}}{D_n}.$$



Three-term recurrence and the Jacobi matrix

Theorem (Three-term recurrence; Favard's converse (1935))

$$xp_n = p_{n+1} + b_n p_n + a_n p_{n-1}, \qquad a_n = \frac{h_n^2}{h_{n-1}^2} > 0, \quad b_n \in \mathbb{R}.$$

Define the orthonormal polynomials $\phi_n = p_n / h_n$. Then

$$x\phi_n = \sqrt{a_{n+1}}\,\phi_{n+1} + b_n \phi_n + \sqrt{a_n}\,\phi_{n-1}.$$

Multiplication by x has a symmetric tridiagonal Jacobi matrix:

$$J = \begin{pmatrix} b_0 & \sqrt{a_1} & 0 & \cdots \\ \sqrt{a_1} & b_1 & \sqrt{a_2} & \ddots \\ 0 & \sqrt{a_2} & b_2 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}.$$



Christoffel–Darboux

The ordinary Christoffel–Darboux kernel is the polynomial sum

$$K_N^{\mathrm{OP}}(x,y) = \sum_{k=0}^{N-1} \frac{p_k(x)p_k(y)}{h_k^2}.$$

Theorem (Christoffel (1858), Darboux (1878); cf. Chebyshev (1855))

$$K_N^{\mathrm{OP}}(x,y) = \frac{p_N(x)p_{N-1}(y) - p_{N-1}(x)p_N(y)}{h_{N-1}^2(x-y)}.$$

It has the *reproducing-kernel property*: for every polynomial f of degree $< N$,

$$\int_{\mathbb{R}} K_N^{\mathrm{OP}}(x,y) f(y)\,\mathrm{d}\mu(y) = f(x).$$



Some applications of orthogonal polynomials

- Gaussian quadrature and Padé approximation
- Spectral methods for differential equations
- Moment problems and logarithmic potential theory
- Jacobi operators and spectral theory
- Hankel determinants and Toda flows



Some applications of orthogonal polynomials

- Gaussian quadrature and Padé approximation
- Spectral methods for differential equations
- Moment problems and logarithmic potential theory
- Jacobi operators and spectral theory
- Hankel determinants and Toda flows
- **Random matrices**



From a Gaussian matrix to Hermite polynomials

The Gaussian unitary ensemble

Let $H = H^*$ be an $N \times N$ Hermitian matrix, with independent real coordinates

$$H_{11}, \dots, H_{NN} \in \mathbb{R}, \qquad \operatorname{Re} H_{ij}, \operatorname{Im} H_{ij} \in \mathbb{R} \quad (i < j).$$

Definition (Gaussian unitary ensemble)

$$\mathrm{d}\mathbb{P}_N(H) = \frac{1}{\mathcal{Z}_N} e^{-\frac{1}{2}\operatorname{Tr} H^2} \prod_{i=1}^{N} \mathrm{d}H_{ii} \prod_{1 \le i < j \le N} \mathrm{d}\operatorname{Re} H_{ij}\, \mathrm{d}\operatorname{Im} H_{ij}.$$

The partition function is

$$\mathcal{Z}_N = \int_{H=H^*} e^{-\frac{1}{2}\operatorname{Tr} H^2}\, \mathrm{d}H = (2\pi)^{N/2} \pi^{N(N-1)/2}.$$



Diagonalizing the matrix: the eigenvalue measure

Diagonalize the matrix H:

$$H = U \operatorname{diag}(\lambda_1, \ldots, \lambda_N) U^*.$$

Theorem (Weyl integration formula (H. Weyl, 1925–26))

The change of variables has Jacobian

$$\mathrm{d}H = C_N \underbrace{\prod_{i<j} (\lambda_i - \lambda_j)^2}_{\Delta(\boldsymbol{\lambda})^2} \, \mathrm{d}\boldsymbol{\lambda} \, \mathrm{d}\nu(U).$$

Integrating over the angular variables U gives the joint eigenvalue measure:

$$\mathrm{d}\mathbb{P}_N^{\mathrm{eig}}(\boldsymbol{\lambda}) = \frac{1}{\widehat{Z}_N} \Delta(\boldsymbol{\lambda})^2 \prod_{i=1}^{N} w(\lambda_i) \, \mathrm{d}\boldsymbol{\lambda}, \qquad \mathrm{d}\mu(x) = w(x)\,\mathrm{d}x, \quad w(x) = e^{-x^2/2}.$$

Heine's identity: the mean characteristic polynomial

Theorem (Heine's integral identity)

For a unitary polynomial ensemble with $\mathrm{d}\mu(x) = w(x)\,\mathrm{d}x$,

$$\begin{aligned}\mathbb{E}\det(zI - H) &= \frac{1}{\widehat{Z}_N}\int_{\mathbb{R}^N}\left(\prod_{i=1}^{N}(z-\lambda_i)\right)\Delta(\boldsymbol{\lambda})^2\prod_{i=1}^{N}w(\lambda_i)\,\mathrm{d}\boldsymbol{\lambda}\\ &= p_N(z).\end{aligned}$$

Here p_N *is the monic orthogonal polynomial with respect to* μ.

For GUE, $\mathrm{d}\mu(x) = e^{-x^2/2}\,\mathrm{d}x$, so $p_N(x) = \mathrm{He}_N(x)$, the monic (probabilists') Hermite polynomial.



The joint eigenvalue law is a kernel determinant

The CD kernel K_N^{OP} acts with respect to $\mathrm{d}\mu = w\,\mathrm{d}x$. Relative to Lebesgue measure, the kernel is

$$\mathcal{K}_N(x,y) = \sqrt{w(x)w(y)}\,K_N^{\mathrm{OP}}(x,y).$$

Theorem (Determinantal OP ensemble)

The joint measure of the N randomly ordered eigenvalues is

$$\mathrm{d}\mathbb{P}_N(\lambda_1,\ldots,\lambda_N) = \frac{1}{N!}\det\left[\mathcal{K}_N(\lambda_i,\lambda_j)\right]_{i,j=1}^{N}\mathrm{d}\lambda_1\cdots\mathrm{d}\lambda_N.$$



Correlation functions

The m-point correlation function counts *ordered selections of m distinct eigenvalues*:

$$\rho_m(x_1, \dots, x_m) = \frac{N!}{(N-m)!} \int_{\mathbb{R}^{N-m}} f_N(x_1, \dots, x_N)\, \mathrm{d}x_{m+1} \cdots \mathrm{d}x_N.$$

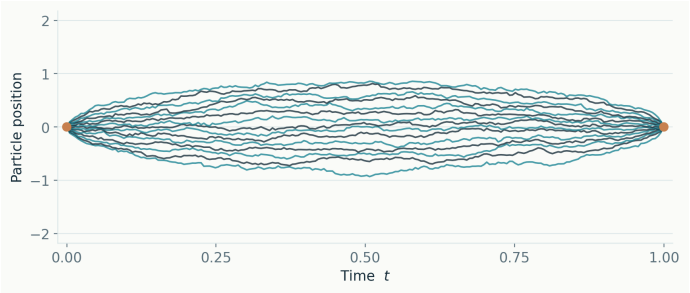
For a determinantal ensemble,

$$\rho_m(x_1, \dots, x_m) = \det[\mathcal{K}_N(x_i, x_j)]_{i,j=1}^m.$$

Interpretation. The functions ρ_m encode the joint statistics of the eigenvalues: $\rho_1(x)\ \mathrm{d}x$ is the expected number of eigenvalues near x, while ρ_m for $m \geq 2$ describes their m-point correlations.



Non-intersecting Brownian bridges and GUE



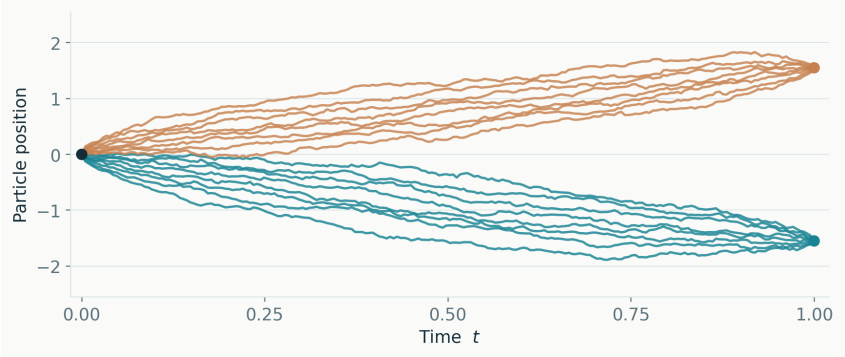
Theorem (Fixed-time GUE correspondence; Dyson (1962), later bridge formulations)

For N nonintersecting Brownian bridges starting and ending at 0 on $[0,1]$, the positions at time t have the eigenvalue law of

$$\widehat{H}_t = \sqrt{t(1-t)}\, H, \qquad H \sim \mathrm{GUE}.$$

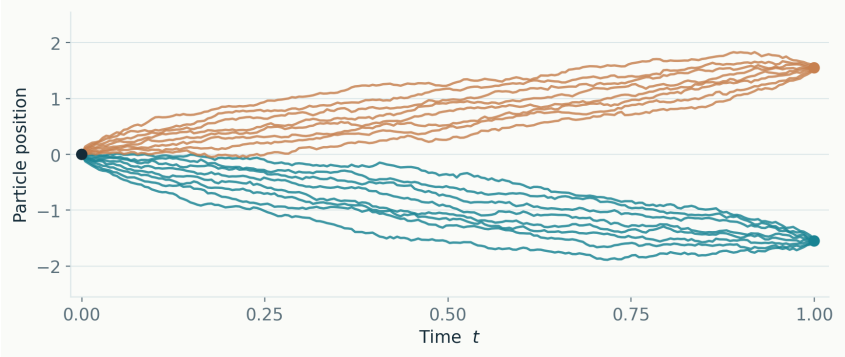
Several endpoints and multiple orthogonality

The next question: several terminal positions



All N paths start at 0, but n_j paths end at a_j ($n_1 + \cdots + n_r = N$).

The next question: several terminal positions



All N paths start at 0, but n_j paths end at a_j ($n_1 + \cdots + n_r = N$).

Questions: Is there still a matrix model? Which orthogonal polynomials arise?

The external-source Gaussian matrix ensemble

Theorem (One-start/many-endpoint correspondence; see van Moerbeke (2007))

Write the terminal-position matrix as

$$A = \operatorname{diag}(\underbrace{a_1, \dots, a_1}_{n_1}, \dots, \underbrace{a_r, \dots, a_r}_{n_r}).$$

At $0 < t < 1$, the bridge positions have the eigenvalue law of

$$\widehat{H}_t = tA + \sigma_t H, \qquad \sigma_t = \sqrt{t(1-t)}, \quad H \sim \text{GUE}.$$

For a fixed time t, rescale by σ_t^{-1} and absorb the factor t/σ_t into the source matrix. We denote this rescaled source again by A:

$$\boxed{X = A + H, \qquad H \sim \text{GUE}.}$$



From the matrix integral to several weights

Theorem (Harish-Chandra (1957), Itzykson–Zuber (1980); Bleher–Kuijlaars (2004))

Let A have distinct eigenvalues a_j, with multiplicities n_j. The Gaussian external-source ensemble

$$\mathrm{d}\mathbb{P}(X) \propto e^{-\operatorname{Tr}(X^2/2 - AX)}\,\mathrm{d}X$$

has joint eigenvalue measure proportional to

$$\Delta(\boldsymbol{x})\,\det\left[x_i^k w_j(x_i)\right]_{\substack{1\le i\le N \\ (j,k):0\le k<n_j}}\mathrm{d}\boldsymbol{x}, \qquad w_j(x) = e^{-x^2/2 + a_j x}.$$



From the matrix integral to several weights

Theorem (Harish-Chandra (1957), Itzykson–Zuber (1980); Bleher–Kuijlaars (2004))

Let A have distinct eigenvalues a_j, with multiplicities n_j. The Gaussian external-source ensemble

$$\mathrm{d}\mathbb{P}(X) \propto e^{-\operatorname{Tr}(X^2/2 - AX)}\,\mathrm{d}X$$

has joint eigenvalue measure proportional to

$$\Delta(\boldsymbol{x})\,\det\left[x_i^k w_j(x_i)\right]_{\substack{1\le i\le N \\ (j,k):0\le k<n_j}}\,\mathrm{d}\boldsymbol{x}, \qquad w_j(x) = e^{-x^2/2 + a_j x}.$$

What about the orthogonal polynomials?



Multiple orthogonal polynomials, types I and II

Fix $r \geq 1$ and $N \geq 1$. Let $\boldsymbol{n} = (n_1, \ldots, n_r)$ with $n_j \geq 0$ and $|\boldsymbol{n}| = N$, and let μ_j $(j = 1, \ldots, r)$ be positive measures on $\mathbb{R}$ with $\mathrm{d}\mu_j = w_j \, \mathrm{d}x$.

Type II (monic polynomial)

$$\deg P_{\boldsymbol{n}} = N, \qquad \int P_{\boldsymbol{n}}(x) x^k \, \mathrm{d}\mu_j(x) = 0, \quad 0 \leq k < n_j.$$

Type I (dual linear form)

$$Q_{\boldsymbol{n}}(x) = \sum_{j=1}^{r} A_{\boldsymbol{n},j}(x) w_j(x), \quad \deg A_{\boldsymbol{n},j} \leq n_j - 1,$$

$$\int x^k Q_{\boldsymbol{n}}(x) \, \mathrm{d}x = \delta_{k,N-1}, \quad 0 \leq k < N.$$

Where MOP came from: Hermite–Padé approximation

Hermite (1873): simultaneous rational approximation of exponential functions in the proof of transcendence of e. For Laurent series $f_1, \dots, f_r$ at infinity, seek one common denominator:

$$f_1(z) \sim \frac{R_1(z)}{P_{\boldsymbol{n}}(z)}, \quad \dots, \quad f_r(z) \sim \frac{R_r(z)}{P_{\boldsymbol{n}}(z)}.$$

Definition (Type-II Hermite–Padé approximation)

$$P_{\boldsymbol{n}}(z) f_j(z) - R_j(z) = O(z^{-n_j-1}), \quad z \to \infty,$$

where $P_{\boldsymbol{n}}$ is monic of degree $N = |\boldsymbol{n}|$ and $\deg R_j \le N - 1$.



The Hermite–Padé denominator is a MOP

For Cauchy transforms $f_j(z) = \int_{\mathbb{R}} (z-x)^{-1} \, \mathrm{d}\mu_j(x)$, taking R_j as the polynomial part of $P_{\boldsymbol{n}} f_j$ gives

$$P_{\boldsymbol{n}}(z) f_j(z) - R_j(z) = \int_{\mathbb{R}} \frac{P_{\boldsymbol{n}}(x)}{z-x} \, \mathrm{d}\mu_j(x).$$

Theorem (Hermite–Padé $\longleftrightarrow$ multiple orthogonality)

The approximation orders are equivalent to

$$\int_{\mathbb{R}} x^k P_{\boldsymbol{n}}(x) \, \mathrm{d}\mu_j(x) = 0, \quad 0 \le k < n_j.$$



Several weights: a bordered mixed Hankel determinant

For r measures, let $m_k^{(j)} = \int_{\mathbb{R}} x^k \, \mathrm{d}\mu_j(x)$ and $\boldsymbol{n} = (n_1, \ldots, n_r)$, $N = |\boldsymbol{n}|$.

$$P_{\boldsymbol{n}}(z) = \frac{1}{\widetilde{D}_{\boldsymbol{n}}} \det \begin{pmatrix} m_0^{(1)} & \cdots & m_{n_1-1}^{(1)} & \cdots & m_0^{(r)} & \cdots & m_{n_r-1}^{(r)} & 1 \\ m_1^{(1)} & \cdots & m_{n_1}^{(1)} & \cdots & m_1^{(r)} & \cdots & m_{n_r}^{(r)} & z \\ \vdots & & \vdots & & \vdots & & \vdots & \vdots \\ m_N^{(1)} & \cdots & m_{N+n_1-1}^{(1)} & \cdots & m_N^{(r)} & \cdots & m_{N+n_r-1}^{(r)} & z^N \end{pmatrix}.$$

$\widetilde{D}_{\boldsymbol{n}}$ is the determinant of the upper-left $N \times N$ moment block.



The random-matrix connection survives

Theorem (Bleher–Kuijlaars (2004))

For the Gaussian external-source ensemble with source eigenvalues a_j of multiplicities n_j,

$$P_{\boldsymbol{n}}(z) = \mathbb{E}\det(zI - X),$$

where $P_{\boldsymbol{n}}$ is the type-II multiple Hermite polynomial.

Theorem (Nearest-neighbor recurrence; Van Assche (2011))

For $1 \le k \le r$, whenever the indicated neighboring MOP are defined,

$$xP_{\boldsymbol{n}} = P_{\boldsymbol{n}+\mathbf{e}_k} + b_{\boldsymbol{n},k}P_{\boldsymbol{n}} + \sum_{j=1}^{r} a_{\boldsymbol{n},j}P_{\boldsymbol{n}-\mathbf{e}_j}.$$

Here $\mathbf{e}_j = (0,\dots,0,1,0,\dots,0)$ is the j-th unit multi-index.

The correlation kernel for multiple Hermite

Let $\{P_k, Q_k\}$ be biorthogonal type-II/type-I families along a multi-index path. Then

$$K_{\boldsymbol{n}}(x,y) = \sum_{k=0}^{N-1} P_k(x) Q_k(y).$$

Theorem (Zinn-Justin (1997); Bleher–Kuijlaars (2004); Daems–Kuijlaars (2004))

The external-source ensemble is determinantal:

$$\rho_m(x_1, \ldots, x_m) = \det\big[K_{\boldsymbol{n}}(x_i, x_j)\big]_{i,j=1}^{m}.$$



Reduction to ordinary orthogonal polynomials

One primary weight and a few perturbing weights

Consider a multi-index $\boldsymbol{n} = (n_0, n_1, \ldots, n_r)$ with $n_0 \gg n_j$. For the primary measure $\mathrm{d}\mu_0 = w_0\,\mathrm{d}x$, the ordinary OP satisfy

$$p_N(z) = \frac{1}{D_N(w_0)} \det \begin{pmatrix} m_0^{(0)} & \cdots & m_{N-1}^{(0)} & 1 \\ m_1^{(0)} & \cdots & m_N^{(0)} & z \\ \vdots & & \vdots & \vdots \\ m_N^{(0)} & \cdots & m_{2N-1}^{(0)} & z^N \end{pmatrix}.$$

For example, take $\boldsymbol{n} = (N-r, 1, \ldots, 1)$, with companion measures $\mathrm{d}\mu_j = w_j\,\mathrm{d}x$. The MOP determinant has extra moment columns:

$$P_{\boldsymbol{n}}(z) = \frac{1}{\widetilde{D}_{\boldsymbol{n}}} \det \begin{pmatrix} m_0^{(0)} & \cdots & m_{N-r-1}^{(0)} & m_0^{(1)} & m_0^{(2)} & \cdots & m_0^{(r)} & 1 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ m_N^{(0)} & \cdots & m_{2N-r-1}^{(0)} & m_N^{(1)} & m_N^{(2)} & \cdots & m_N^{(r)} & z^N \end{pmatrix}.$$



Change the basis: from bordered Hankel to a small matrix

Replace the monomials in the bordered Hankel determinant $\widetilde{D}_{\boldsymbol{n}}$ by the ordinary polynomials p_k for μ_0. Elementary triangular basis changes give

$$\widetilde{D}_{\boldsymbol{n}} \quad \longrightarrow \quad M = \begin{pmatrix} H_0 & M_{12} \\ 0 & M_{22} \end{pmatrix}, \qquad H_0 = \operatorname{diag}(h_0^2, \ldots, h_{N-r-1}^2).$$

Only the *small block* M_{22} remains nontrivial:

$$\dim M_{22} = q = N - (N - r) = r.$$

Thus a fixed number of extra weights produces a fixed-size inverse, regardless of the degree N.



The finite-dimensional data

Let $m = N - r$, and let p_k be monic OP for the reference measure $\mathrm{d}\mu_0 = w_0\,\mathrm{d}x$, with $h_k^2 = \int_{\mathbb{R}} p_k(t)^2\,\mathrm{d}\mu_0(t)$. The reduced matrix and moment vector are

$$(M_{22})_{ij} = \int_{\mathbb{R}} p_{m+i-1}(t)\,\mathrm{d}\mu_j(t), \qquad (\boldsymbol{v}_2)_j = \int_{\mathbb{R}} p_N(t)\,\mathrm{d}\mu_j(t), \quad 1 \le i, j \le r.$$

The reference CD kernel is the *degree-m* kernel, not the full degree-N kernel:

$$K_m^{\mathrm{OP}}(x,y) = \sum_{k=0}^{m-1} \frac{p_k(x)p_k(y)}{h_k^2}, \qquad K_N^{\mathrm{OP}}(x,y) = \sum_{k=0}^{N-1} \frac{p_k(x)p_k(y)}{h_k^2}.$$

The projected weight functions are

$$r_j(y) = w_j(y) - w_0(y) \int_{\mathbb{R}} K_m^{\mathrm{OP}}(t,y)\,\mathrm{d}\mu_j(t).$$



Finite-dimensional reduction of polynomials and kernels

Theorem (Gharakhloo–K.–Miyahara (2026))

Assume that the reduced mixed-moment block M_{22} is invertible. Set $\boldsymbol{p}_2 = (p_{N-r}, \dots, p_{N-1})^T$. Then

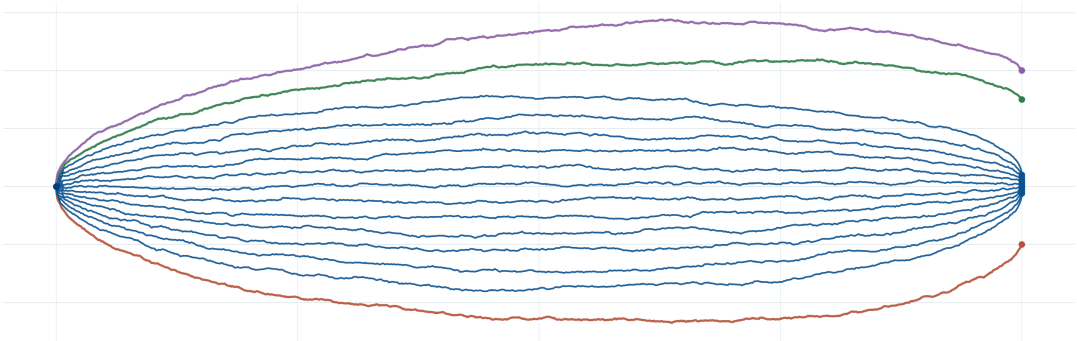
$$\boxed{P_{\boldsymbol{n}}(x) = p_N(x) - \boldsymbol{v}_2^T M_{22}^{-1} \boldsymbol{p}_2(x).}$$

With compatible biorthogonal conventions, the associated kernel satisfies

$$\boxed{K_{\boldsymbol{n}}(x,y) = w_0(y) K_m^{\mathrm{OP}}(x,y) + \boldsymbol{r}(y)^T M_{22}^{-1} \boldsymbol{p}_2(x).}$$

Here $\boldsymbol{v}_2$, $\boldsymbol{r}$ and M_{22} are defined on the preceding slide.

Gaussian external sources: finitely many spikes



For Gaussian weights with external source, $w_0(x) = e^{-x^2/2}$ and $w_j(x) = e^{-x^2/2+a_j x}$. Small-rank perturbations of this kind were studied by Baik (2009), and they are closely related to the *spiked models* of random matrix theory.

The unit circle and escaping zeros

Ordinary orthogonal polynomials on the unit circle

Definition (OPUC)

Let μ be a finite positive measure on $\mathbb{T} = \{z : |z| = 1\}$. The nth *orthogonal polynomial on the unit circle* is the unique monic polynomial $\Phi_n(z) = z^n + \cdots$ satisfying

$$\int_{\mathbb{T}} \Phi_n(z) \bar{z}^k \, \mathrm{d}\mu(z) = \delta_{nk} h_n^2, \qquad k = 0, \ldots, n,$$

where $h_n^2 = \int_{\mathbb{T}} |\Phi_n(z)|^2 \, \mathrm{d}\mu(z) > 0$.

Let $c_\ell = \int_{\mathbb{T}} z^\ell \, \mathrm{d}\mu(z)$ be the Fourier moments.

Theorem (Toeplitz determinant formula)

With $T_n = \det(c_{j-k})_{j,k=0}^{n-1}$,

$$\Phi_n(z) = \frac{1}{T_n} \det \begin{pmatrix} c_0 & c_{-1} & \cdots & c_{1-n} & 1 \\ c_1 & c_0 & \cdots & c_{2-n} & z \\ \vdots & \vdots & & \vdots & \vdots \\ c_n & c_{n-1} & \cdots & c_1 & z^n \end{pmatrix}.$$

Multiple orthogonality on the unit circle

Let $\mu_0, \dots, \mu_r$ be circle measures, $\boldsymbol{n} = (n_0, \dots, n_r)$ and $N = |\boldsymbol{n}|$. A monic MOPUC satisfies

$$\int_{\mathbb{T}} \Phi_{\boldsymbol{n}}(z) \bar{z}^k \, \mathrm{d}\mu_j(z) = 0, \quad 0 \le k < n_j, \quad 0 \le j \le r.$$

Write $c_\ell^{(j)} = \int_{\mathbb{T}} z^\ell \, \mathrm{d}\mu_j(z)$. The mixed Toeplitz determinant is

$$\Phi_{\boldsymbol{n}}(z) = \frac{1}{\widetilde{T}_{\boldsymbol{n}}} \det \begin{pmatrix} c_0^{(0)} & \cdots & c_{1-n_0}^{(0)} & c_0^{(1)} & \cdots & c_{1-n_1}^{(1)} & c_0^{(2)} & \cdots & c_{1-n_r}^{(r)} & 1 \\ c_1^{(0)} & \cdots & c_{2-n_0}^{(0)} & c_1^{(1)} & \cdots & c_{2-n_1}^{(1)} & c_1^{(2)} & \cdots & c_{2-n_r}^{(r)} & z \\ \vdots & & \vdots & \vdots & & \vdots & \vdots & & \vdots & \vdots \\ c_N^{(0)} & \cdots & c_{N-n_0+1}^{(0)} & c_N^{(1)} & \cdots & c_{N-n_1+1}^{(1)} & c_N^{(2)} & \cdots & c_{N-n_r+1}^{(r)} & z^N \end{pmatrix}.$$

$\widetilde{T}_{\boldsymbol{n}}$ is the determinant of the first N rows and N moment columns.



Ordinary OPUC: the simplest possible reference measure

On $\mathbb{T}$, choose Haar measure $\mathrm{d}\mu(e^{i\theta}) = \mathrm{d}\theta/(2\pi)$. Then

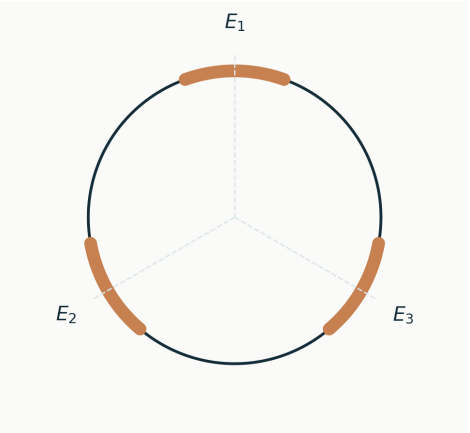
$$\Phi_N(z) = z^N, \qquad K_N(z,\zeta) = \sum_{k=0}^{N-1} z^k \bar{\zeta}^k = \frac{1-(z\bar{\zeta})^N}{1-z\bar{\zeta}}.$$

Theorem (Fejér zero localization)

For a positive measure on $\mathbb{T}$ whose support contains more than N points, all zeros of Φ_N lie strictly inside the unit disk.



Add arc-supported weights: a simple MOPUC model



$$w_0 \equiv 1, \qquad w_j = \mathbf{1}_{E_j}, \quad j = 1, \ldots, r.$$

Choose multi-index $(N - r, 1, \ldots, 1)$. The $N - r$ Haar conditions force the lower coefficients to vanish:

$$\Phi_N(z) = z^N + \sum_{k=1}^{r} c_k z^{N-k}.$$

MOPUC: both polynomial and kernel reductions

Theorem (Gharakhloo–K.–Miyahara (2026))

For an OPUC reference basis and invertible reduced mixed-moment block,

$$\Phi_{\boldsymbol{n}}(z) = \Phi_N^{\mathrm{OPUC}}(z) - \boldsymbol{v}_2^T M_{22}^{-1} \boldsymbol{\Phi}_2(z),$$

$$K_{\mathrm{MOPUC}}(z,\zeta) = K_{\mathrm{OPUC}}(z,\zeta) + \boldsymbol{\pi}(\zeta)^T M_{22}^{-1} \boldsymbol{\phi}_2(z).$$



Equal arcs give a regular polygon of zeros

Take r arcs of length Δ centered at $x_j = x_1 e^{2\pi i(j-1)/r}$. Their Fourier coefficients are

$$a_k = \frac{\sin(k\Delta/2)}{\pi k} \quad (k \neq 0), \qquad a_0 = \frac{\Delta}{2\pi}.$$

Proposition (Symmetric arcs; Gharakhloo–K.–Miyahara (2026))

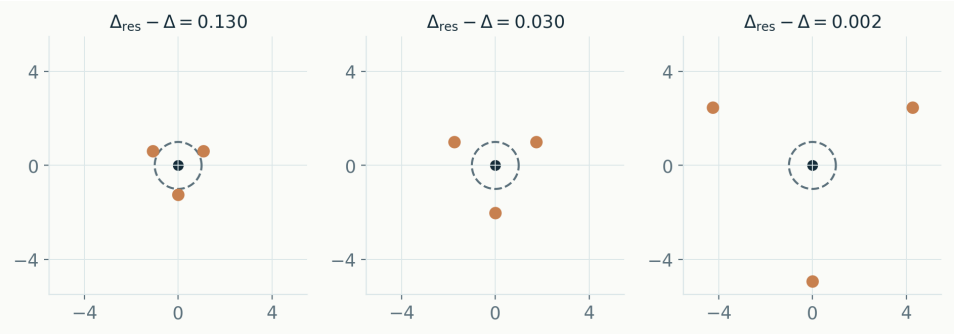
If the mixed moment matrix is invertible, then

$$\Phi_N(z) = z^{N-r}\left(z^r - \frac{a_N}{a_{N-r}} x_1^r\right).$$

There are $N - r$ zeros at 0, while the remaining r zeros form a regular polygon of radius $|a_N/a_{N-r}|^{1/r}$.



Resonance: the polygon escapes



Example: $N = 8, r = 3$. The three nonzero zeros move away from the unit circle as the arc length approaches a resonance.

What we have not discussed

1. **Large-N limits.** When does the zero-counting measure $\nu_N = \frac{1}{N}\sum_{P_N(\zeta)=0}\delta_\zeta$ have the same limit as $\mu_N^{\text{eig}} = \frac{1}{N}\sum_{j=1}^{N}\delta_{\lambda_j}$?
2. **Riemann–Hilbert analysis.** A systematic method for strong asymptotics of OP/MOP.
3. **Local universality.** Bulk, soft edges, and spikes; sine, Airy, Pearcey and Painlevé structures.
4. **Non-Gaussian potentials.** More general $V(x)$ and one-cut/two-cut phase transitions (Bleher–Kuijlaars).
5. **More endpoints at both times.** Two-HCIZ matrix models for Brownian bridges with several start and end positions.
6. **Zeros on the circle.** Large-degree MOPUC zero measures, escape exponents, and resonance geometry.





Thank you

Selected references

Classical OP, approximation and probability

- G. Szegő, *Orthogonal Polynomials* (1939).
- Ch. Hermite, *Sur la fonction exponentielle* (1873).
- J. Favard, *Sur les polynômes de Tchebicheff* (1935).
- M. L. Mehta, *Random Matrices*, 3rd ed. (2004).
- F. J. Dyson, *A Brownian-motion model for the eigenvalues of a random matrix*, JMP (1962).
- M. Adler, J. Delépine and P. van Moerbeke, *Dyson's nonintersecting Brownian motions with a few outliers*, CPAM (2009).
- A. Hardy, *Average characteristic polynomials of determinantal point processes*, AIHP (2015).

External sources, MOP and reduction

- P. Zinn-Justin, *Universality of correlation functions ... in an external field*, CMP (1997).
- P. Bleher and A. Kuijlaars, *Random matrices with external source and multiple OP*, IMRN (2004).
- E. Daems and A. Kuijlaars, *A Christoffel–Darboux formula for MOP*, JAT (2004).
- J. Baik, *On the Christoffel–Darboux kernel ... with external source* (2009).
- W. Van Assche, *Nearest neighbor recurrence relations for MOP*, JAT (2011).
- R. Gharakhloo, M. Kosmakov and K. Miyahara, *Reduction of Multiple Orthogonal Polynomials to Standard Orthogonal Polynomials* (2026).

